

TANU SINGH

Software Engineer — AI/ML Engineer

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Education

B.Tech in Computer Science and Engineering

Vellore Institute of Technology, Bhopal

2023 – 2027

CGPA: 8.81 / 10

Technical Skills

- **Languages:** Java, Python, SQL, TypeScript
- **AI/ML & GenAI:** Machine Learning, Deep Learning, NLP, Transformers, LLMs, RAG, Embeddings, Semantic Search, Prompt Engineering, Google Gemini API
- **Backend:** Spring Boot, Spring Security, Spring Data JPA, Hibernate, FastAPI, REST APIs, Microservices, Multithreading, Concurrency, Java Virtual Threads
- **Frontend:** React, Tailwind CSS, Vite, Streamlit
- **Databases:** MySQL, PostgreSQL, Redis, Qdrant (Vector DB), Firebase
- **Cloud & DevOps:** AWS, Google Cloud, Docker, GitHub Actions, CI/CD, Linux, Apache Kafka, Maven, Git, Postman, JWT
- **Testing:** JUnit, Mockito, Unit Testing, Integration Testing, Concurrency Testing
- **Core CS:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design, Design Patterns, Competitive Programming

Experience

Backend Engineer Intern

Rasient Technohub Pvt. Ltd.

Jan 2026 – March 2026

Remote

- Developed scalable **REST APIs** using **Java**, **Spring Boot**, and **Microservices** architecture.
- Integrated backend systems with **AI/ML services** to power intelligent features, and implemented **event-driven architecture** using **Apache Kafka** for real-time data flow.
- Optimized performance using **Redis** caching, improving API response time and overall system scalability.
- Deployed and managed containerized services using **Docker** and **CI/CD pipelines (GitHub Actions)** on **Linux-based** cloud platforms (**AWS** and **Google Cloud**), ensuring reliable releases.
- Collaborated in an **Agile** team through **sprint planning** and **code reviews**.

Projects

TimeForge - AI-Powered Productivity App

Feb 2026

- Built an **AI-powered productivity and study management platform** using **React**, **TypeScript**, **Tailwind CSS**, **Vite**, **Java**, and **Spring Boot**, applying **prompt engineering** for context-aware study guidance.
- Integrated the **Google Gemini API (LLM)** through the Spring Boot backend to deliver **AI-generated summaries, explanations**, and personalized study guidance based on user syllabus, tasks, and exam data.
- Developed **RESTful APIs** using **Spring Boot**, **Spring Data JPA**, and **H2** for persistent management of **tasks, timetables, syllabus modules, exams**, and productivity records, replacing the previous client-only storage approach.
- Implemented **JWT-based authentication with Spring Security** and structured the frontend using **React Context API, custom hooks**, and API services for synchronized client-server state management.

Order Matching Engine

April 2026

- Engineered a Java-based **order matching engine** implementing **price-time priority (FIFO)** for deterministic and fair trade execution.
- Designed an in-memory **order book** using **TreeMap**-based price levels, enabling **$O(\log n)$** order insertion and efficient best-price retrieval.
- Implemented trade matching with support for **partial fills, order cancellation**, and generation of **execution events** for completed trades.
- Applied **ReentrantLock**-based synchronization with an asynchronous **producer-consumer event pipeline** to maintain consistency during **concurrent order processing**.

RAG App - Retrieval-Augmented Generation

April 2026

- Developed a **Retrieval-Augmented Generation (RAG)** application using **FastAPI**, **Qdrant**, and the **Google Gemini API** for querying uploaded PDFs and generating context-grounded answers.
- Built an async ingestion pipeline with **Inngest** covering **pypdf text extraction, sliding-window chunking (4000/500 overlap), 3072-dim Gemini embeddings**, and **top-10 cosine-similarity search** in Qdrant.
- Exposed **REST APIs (/ingest, /query)** using **FastAPI** and built a **Streamlit** interface for PDF upload and question answering.

Certifications & Achievements

- **IBM Generative AI Certification** (Apr 2026): Covered **prompt engineering, LLM workflows, transformers, and AI application development**.
- Solved **400+ DSA and competitive programming problems** across Codeforces, LeetCode, and CodeChef; ranked **1st in college** in multiple contests.
- Led a team of 4 to win the **Codeverse Hackathon** among **250+ participating teams**.